Problem Set 2

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market design
Q3

Part 1 : Market Setup

Python Code

Part 2: Matching Mechanisms

(i) Deferred Acceptance (DA) (Student → School):

- S1: i3, i5, i8, i12, i4, i18
- S2: i7, i9, i10, i13, i15, i16
- S3: i1, i2, i4, i6, i11, i17

(ii) Immediate Acceptance (IA):

- S1: i5, i8, i12, i14, i17, i18
- S2: i7, i9, i10, i13, i15, i16
- S3: i1, i2, i3, i4, i6, i11

(iii) TTC:

- S1 : i5, i7, i8, i12, i14, i18
- S2: i9, i10, i13, i15, i16, i17
- S3: i1, i2, i3, i4, i6, i11

All three mechanisms give us different allocations in the market.

Table format according to students on next page →

Student	DA	IA	TTC
i1	S3	S3	S3
i2	S3	S3	S3
i3	S1	S3	S3
i4	S3	S3	S3
i5	S1	S1	S1
i6	S3	S3	S3
i7	S2	S2	S1
i8	S1	S1	S1
i9	S2	S2	S2
i10	S2	S2	S2
i11	S3	S3	S3
i12	S1	S1	S1
i13	S2	S2	S2
i14	S1	S1	S1
i15	S2	S2	S2
i16	S2	S2	S2
i17	S3	S1	S2
i18	S1	S1	S1

Part 3: Efficiency Analysis

N = 1000

[1 = top choice
2 = second choice
3 = third choice]

DA, IA and TTC $\mu^{DA}, \mu^{IA}, \mu^{TTC}$

$$\text{AvgRank(mechanism)} = \sum_{t=1}^{1000} \sum_{i=1}^{18} \text{rank}(\mu_t^m)$$

DA assigned : 18
IA assigned : 18
TTC assigned : 18

Results (mechanism avg. student rank)

DA → 1.208611
IA → 1.183722
TTC → 1.193389

Interpretation

A lower average rank is linked to better student outcomes. The order of ranks (lowest to highest) is IA → TTC → DA. IA being the lowest, makes it the best performer for average student ranking.

This is important as a researcher because it helps determine how the choice of mechanism has an effect on student welfare. This stands also when these rankings are generated randomly. DA → Stable, strategy proof ; avg. student rank not maximized. IA → Better avg. rank outcome since high rank students get first priority.